24. Friday May 23, Composition (6.4)

Theorem 1: Let $f: A \to B$ and $g: B \to C$. The following hold:

1) If $g \circ f$ is injective then f is injective.

2) If $g \circ f$ is surjective then g is surjective.

Proof:

1) Suppose $g \circ f$ is injective. We want to prove that f is injective. Thus, let $x, y \in A$ such that $f(x) = f(y)$. We want to show $x = y$. Since $f(x) = f(y)$, we can apply g to get $g(f(x)) = g(f(y))$. By def^n of composition, this means $(g \circ f)(x) = (g \circ f)(y)$. Since $g \circ f$ is injective and $(g \circ f)(x) = (g \circ f)(y)$, it must be $x = y$, as desired.

2) Exercise □

Exercise: Give another proof for 1 by contrapositive.

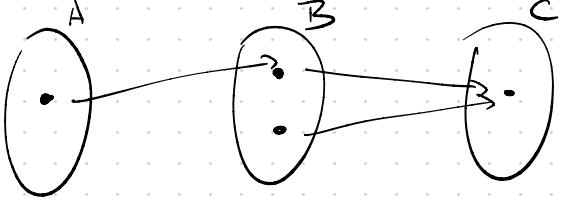
Exploration: Assume $g \circ f$ is surjective.

This says: for every $c \in C$, there is some $a \in A$ such that

$$\underbrace{(g \circ f)(a)}_{g(f(a))} = c$$

So, $\forall c \in C$, $\exists\, b \in B$ (namely $b = f(a)$) such that $g(b) = c$. g surjective!

Can we say more? Let's do some examples

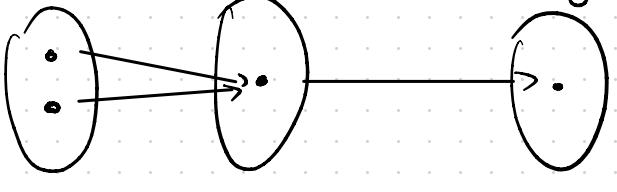


Here $g \circ f$ is surjective,

f is injective, not surjective

and g is not injective

Does f have to be injective?



$g \circ f$ surjective

g injective

f surjective, f not injective

In conclusion. If $g \circ f$ is surjective, then:

- g is surj, may or may not be injective
- f may or may not be inj. or surjective

Exercise: Assume $g \circ f$ is injective. Do a similar exploration.

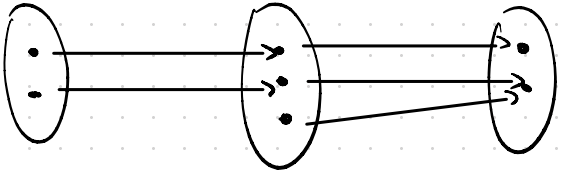
Question: What about bijectivity?

Suppose $g \circ f$ is bijective. Know $g \circ f$ inj $\Rightarrow f$ inj

$g \circ f$ surj $\Rightarrow g$ surj

Turns out f may or may not be surj

g may or may not be inj



f not surjective

g not inj.

Conclusion. Thm 1 is the strongest result we can have, even if we assume $g \circ f$ is bijective.

Theorem 2: Let $f: A \to B$ and $g: B \to C$. The following hold:

1) If f and g are injective, Then $g \circ f$ is injective.

2) If f and g are surjective, Then $g \circ f$ is surjective.

Proof: 1) Exercise.

2) Suppose $f: A \to B$ and $g: B \to C$ are surjections.

We want to prove that $g \circ f: A \to C$ is surjective.

To do this, take an arbitrary $z \in C$. We want to find some $x \in A$ such that $(g \circ f)(x) = z$.

Since g is surjective, $\exists y \in B$ s.t. $g(y) = z$.

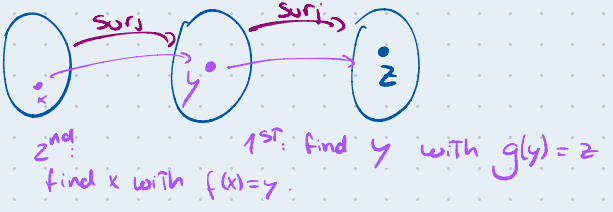
Since f is surjective, $\exists x \in A$ s.t. $f(x) = y$

Then, for this $x \in A$, we have

$$(g \circ f)(x) = g(f(x)) = g(y) = z.$$

Since $z \in C$ was arbitrary, this proves $g \circ f$ is surjective $\square$

Scratch:

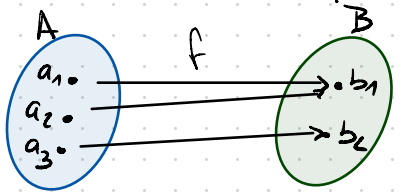


Corollary 3: Let $f: A \to B$ and $g: B \to C$.
If f and g are bijections, Then $g \circ f$ is a bijection

Proof: Exercise!

Intro: Functions as sets.

Motivation: Consider the function



so $f: A \to B$ given by

$f(a_1) = b_1$, $f(a_2) = b_1$, $f(a_3) = b_2$

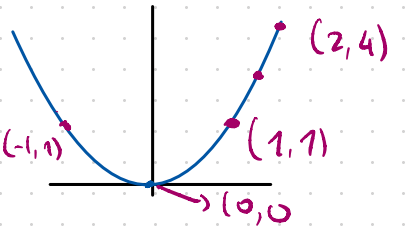
(a_1, b_1), (a_2, b_2), $(a_3, b_2) \in A \times B$

Idea: Codify each of these three defining equalities in an ordered pair $\in A \times B$
If we know these pairs, we can recover f!!
So instead of thinking of $f: A \to B$ as a rule, we can think of it as a subset of $A \times B$:

$$f = \{(a_1, b_1), (a_2, b_1), (a_3, b_2)\}$$

Example: Let $f: \mathbb{R} \to \mathbb{R}$, given by $f(x) = x^2$

We immediately visualize



These pairs look like (x, x^2)

We are just drawing the set of pairs (x, x^2)

So $f = \{(x,y) \in \mathbb{R}^2 \mid y = x^2\}$.

Definition 1: Let $f: A \to B$ be a function.

The ordered pair representation of f (or graph of f) is the subset

$$f = \{(a,b) \in A \times B \mid b = f(a)\} = \{(a, f(a)) \mid a \in A\} \subseteq A \times B$$

Question: Which subsets of $A \times B$ come from some function $A \to B$?

Example: $A = \{a_1, a_2, a_3\}$, $B = \{b_1, b_2\}$

Then $\{(a_1, b_2), (a_2, b_1)\}$ does not represent a function (Don't have output for a_3)

Also $\{(a_1, b_1), (a_2, b_1), (a_3, b_2), (a_3, b_1)\}$ does not represent a function

Because we have two outputs for the input a_3.